

# ZHONGSHENG WANG

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## Summary/Objective

I am currently a second-year PhD student in Computer Science at the University of Auckland, with a research focus on constructing controllable Multimodal Large Language Models (MLLMs). My current research focuses on self-evolving agents and the development of trustworthy multi-agent systems, with a commitment to making practical and impactful contributions to the field of AI.

## Education

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| <b>The University of Auckland</b><br><i>Doctor of Philosophy in Computer Science (Supervisors: Prof. Jiamou Liu and Dr. Qian Liu)</i> | <b>Auckland, New Zealand</b><br>2025.04 - 2026.03 |
| <b>The University of Auckland</b><br><i>Master of Data Science (Supervisor: Prof. Jiamou Liu)</i>                                     | <b>Auckland, New Zealand</b><br>2022.07 - 2024.09 |
| <b>Southwest University</b><br><i>Bachelor of Engineering in Computer Science and Technologies (Advisor: Prof. Li Li)</i>             | <b>Chongqing, China</b><br>2019.09 - 2023.06      |

## Research Interests

- Multimodal Large Language Models (MLLMs)
- Self-evolving AI agents
- Multi-agent systems

## Project Experience

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| <b>BedaiaTi.ai: Agentic AI System for Culturally Grounded Storybook Generation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>2025.04 – 2025.07</b> |
| <ul style="list-style-type: none"><li>• Led the core development of a multi-agent AIGC system for Arabic-context children's storybook generation, coordinating specialized agents for story generation, page-level storyboard planning, character-consistent illustration generation, and language localization.</li><li>• Designed an asynchronous task orchestration workflow based on message queues and developed a reusable model execution control layer, or <i>model harness</i>, to support task scheduling, output validation, logging, and real-time progress feedback.</li><li>• Delivered and deployed the system at bedaiati.ai, enabling an end-to-end user workflow from theme selection and story input to complete storybook generation for the Arabic-speaking market.</li></ul> |                          |

## Industry Experience

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| <b>AI Engineer</b><br><i>Atom Intelligence</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>2024.05 – 2024.08</b><br><i>Remote</i>      |
| <ul style="list-style-type: none"><li>• Built a Chinese-English CRM speech recognition system based on Whisper and LoRA fine-tuning, covering domain adaptation, model deployment, and business system integration; reduced key error-rate metrics by approximately 30% relative to the previous method.</li><li>• Designed an automatic annotation and cleaning workflow for industrial speech data, building a reusable ASR data processing pipeline for training data construction, quality filtering, and iterative model improvement.</li><li>• Participated in experimental validation and deployment optimization of the weakly supervised ASR system, connecting data construction, model fine-tuning, and deployment feedback into a closed-loop workflow for real CRM scenarios.</li></ul> |                                                |
| <b>Data/AI Scientist, Summer Intern</b><br><i>HouGarden Co, Ltd.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>2023.11 – 2024.01</b><br><i>New Zealand</i> |
| <ul style="list-style-type: none"><li>• Built a property-level issue management database for a real estate chatbot by organizing property information, user questions, and question-answering data, providing a structured knowledge base for industry-specific QA systems.</li><li>• Constructed Chinese-English bilingual corpora and a domain terminology database for real estate, and fine-tuned, adapted, and deployed a large language model for official website property information translation, improving stylistic consistency in domain-specific text generation.</li></ul>                                                                                                                                                                                                             |                                                |

## Teaching & Academic Talks

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### Graduate Teaching Assistant

2023.07 – Present

*The University of Auckland*

*Auckland, New Zealand*

- Supported undergraduate and postgraduate computer science courses by answering student questions, marking assignments and tests, and providing feedback on programming and mathematical reasoning tasks.
- Assisted with courses across artificial intelligence, intelligent agents, software engineering theory, and mathematics for computer science.
- Selected courses:** COMPSCI 767 Intelligent Software Agents (2026 S1); COMPSCI 713 AI Fundamentals (2026 S1); COMPSCI 367 Artificial Intelligence (2023 S2, 2025 S2); COMPSCI 761 Advanced Topics in Artificial Intelligence (2024 S2); SOFTENG 282 Software Engineering Theory (2025 S1); COMPSCI 120 Mathematics for Computer Science (2025 S1).

### Guest Lectures and Invited Talk

2024 – 2026

- Text-to-Image Diffusion Models: Limits and Control**, guest lecture for COMPSCI 714 AI Architecture and Design, University of Auckland, May 2026.
- Reinforcement Learning for LLMs: From Post-Training to Skills and Agentic Systems**, guest lecture for COMPSCI 713 AI Fundamentals, University of Auckland, May 2026.
- Creative Intelligence: Applications of Large Language Models in Data Generation and Reasoning**, invited talk, University of Electronic Science and Technology of China, Chengdu, China, December 2024.

## Academic Services

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**Conference Reviewer:** ACM MM 2026; ICONIP 2026; ACL Rolling Review 2026 January; AAAI 2025; ACM MM 2025; COLM 2025; IJCNN 2025; ICONIP 2025; ACL Rolling Review 2025 May, July, October; IJCNN 2024; ICONIP 2024.

**Journal Reviewer:** IEEE Transactions on Fuzzy Systems, 2025.

**Workshop Reviewer:** Reasoning and Planning for Large Language Models @ ICLR 2025.

## Awards & Honors

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**Best Paper Award**, AAMAS, 2026.

**Best Multimedia Award**, ACM Multimedia Asia, 2025.

**Master's Dissertation in First-Class**, University of Auckland, 2024.

**University-level Scholarship**, Southwest University, 2020, 2021, 2022.

## Publications

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### Conference & Journal Papers

\* means equal contribution

Lin Z, **Wang Z**, Liu Q, Zhang X, and Liu J. Narratology Meets Text-to-Image: A Survey of Consistency in AI Generated Storybook Illustrations. *Artificial Intelligence Review*, 2026.

Li M\*, **Wang Z\***, Li H, and Liu J. R-Debater: Retrieval-Augmented Debate Generation through Argumentative Memory. Accepted by *The 25th International Conference on Autonomous Agents and Multiagent Systems*, AAMAS 2026. **Best Paper Award**

Deng S, **Wang Z**, Mao R, Ciprian D G, and Liu J. CLER: Improving Multimodal Financial Reasoning by Cross-MLLM Error Reflection. Accepted by *The 40th AAAI Conference on Artificial Intelligence*, AAAI 2025.

Li X, Ni L, Wang X, Tang Y, Li R, Liu J, and **Wang Z**. LLM-based Business Process Models Generation from Textual Descriptions. Accepted by *The 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, IJCNLP-AACL 2025.

**Wang Z**, Lin M, Lin Z, Shakib Y, Liu Q, and Liu J. CharCom: Composable Identity Control for Multi-Character Story Illustration. Accepted by *The 7th ACM International Conference on Multimedia in Asia*, ACM MM Asia 2025. **Best Multimedia Award**

**Wang Z**, Wang S, Wang J, Liang Y, Zhang Y, and Liu J. Weak Supervision Techniques towards Enhanced ASR Models in Industry-level CRM Systems. *International Conference on Neural Information Processing*, ICONIP 2024.

**Wang Z**, Liu J, Bao Q, Rong H, and Zhang J. ChatLogic: Integrating Logic Programming with Large Language Models for Multi-Step Reasoning. *International Joint Conference on Neural Networks*, IJCNN 2024.

Qi Q, Ni L, **Wang Z**, Zhang L, Liu J, and Witbrock M. Epic-Level Text Generation with LLM through Auto-Prompted Reinforcement Learning. *International Joint Conference on Neural Networks*, IJCNN 2024.

## **Preprint & Workshop Papers**

Xiao X, Shen S, Bao Q, Rong H, Liu K, **Wang Z**, and Liu J. CoRA: Optimizing Low-Rank Adaptation with Common Subspace of Large Language Models. Arxiv pre-print